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BCA (Sem-III) – Software

Testing (304) SEC-I

2022

Time : 3 hours

Full Marks : 70

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Answer all sections as directed.

Section-A

1. Choose the correct alternative for each of the following questions : $2 \times 10 = 20$

(a) Identify the incorrect testing technique.

(i) Integration testing

~~(ii) Collaboration testing~~

(iii) System testing

(iv) Unit testing

(b) By whom is unit testing done ?

~~(i) Users~~

(ii) Customers

(iii) Developers (iv) None

(c) In which of the following categories can white box testing be classified ?

(i) Design based testing

(ii) Structural testing

(iii) Error guessing technique

(iv) None

(d) What is cyclomatic complexity ?

(i) Black box testing

(ii) White box testing

(iii) Gray box testing

(iv) All of these

(e) To which boundary value analysis belongs :

(i) Black box testing

(ii) White box testing

(iii) Both (i) and (ii)

(iv) None

(f) What are various testing levels ?

(i) Unit testing

(ii) System testing

(iii) Integration testing

(iv) All of these

- (g) A program having high cyclomatic complexity is more likely to be :
- (i) Small (ii) Difficult to test
 - (iii) Difficult to write (iv) Large
- (h) Black box testing is also known as :
- (i) Behavioral testing
 - (ii) Data flow testing
 - (iii) Graph based testing
 - (iv) Loop testing
- (i) System testing is a :
- (i) Gray box testing
 - (ii) White box testing
 - (iii) Black box testing
 - (iv) Both (ii) & (iii)
- (j) To notify the developer which bugs need to be fixed first is termed as :
- (i) Scalability (ii) Fixability
 - (iii) Traceability (iv) Priority

Section-B

Short Answer type Questions

Answer any four questions of the following :

5×4=20

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2. What do you mean by boundary value analysis ?
3. What is verification and validation in software testing ?
4. What is the difference between functional and non-functional testing ?
5. Explain the objective of software testing.
6. Define software testing.
7. Explain Graph matrices with example.
8. Explain program graph.
9. What is white box testing ?

Section-C

Long Answer type Questions

Answer any two questions of the following :

15×2=30

10. Explain V-shaped software life cycle model.
11. What is coverage and what are the different types of coverage technique ?
12. Explain types of testing strategies.
13. Explain different types of testing methods.

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